

Current-Normalized Lookup LIF Units: A Boundary Study for Spiking Transformer Neuron Replacement

Anonymous submission

Abstract

Lookup tables can reproduce low-precision spiking neuron updates, but our experiments show that learning or calibrating such replacements is sharply setting-dependent. In a pre-registered QKFormer CIFAR-100 proof of concept, a 6-bit trainable Dense LUT-IF unit degraded Acc@1 from 81.02% to 78.42%, underperforming both a post-hoc dense transition table and a quantized arithmetic LIF control. We therefore test CNL-LUT-LIF, a current-normalized lookup LIF unit that separates lookup-current quantization, temporal-channel current normalization, and an exportable fixed LIF transition table. A fixed comparison matrix across retained CIFAR-10/100, $T = 1/4$, seed-42 QKFormer checkpoints rejects a broad superiority claim: only CIFAR-100 $T = 4$ passes the registered per-run gate, improving post-hoc lookup from 79.55% to 79.77% Acc@1, while CIFAR-10 rows tie near-lossless post-hoc lookup and CIFAR-100 $T = 1$ exposes a severe 6-bit all-LIF replacement drop. A fixed 8-bit follow-up reduces the CIFAR-100 $T = 1$ drop from 6.79 to 0.46 points while still tying post-hoc lookup, localizing that failure to transition precision rather than CNL-LUT-LIF superiority. The resulting contribution is a boundary and diagnostic study: finite LIF lookup is executable, dense transition learning fails, and current normalization is useful in a specific dynamic regime but is not a general lookup-neuron solution.

Introduction

Spiking neural networks (SNNs) replace conventional activations with temporally evolving membrane states and sparse spike events. Recent spiking Transformers demonstrate that this stateful computation can support competitive visual representations while preserving explicitly discrete internal signals (Zhou et al. 2023; Yao et al. 2023; Zhou et al. 2024; Shi, Hao, and Yu 2024). Yet the neuron itself is usually treated as a fixed numerical primitive. Network weights are optimized for a task, whereas the state update remains a hand-designed recurrence such as integrate-and-fire (IF) or leaky integrate-and-fire (LIF). Learnable-neuron work relaxes selected continuous parameters, including the membrane time constant (Fang et al. 2020), but still evaluates an explicit arithmetic recurrence at every time step.

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This paper studies a different question: *where should lookup be inserted so that a spiking block remains dynamically compatible with its trained network?* The question is motivated by two observations. First, the state, input current, spike, and next state of a low-precision spiking neuron form a finite interface. Second, LUT-oriented neural models show that learned discrete functions can be organized under strong structural constraints (Wang et al. 2019). A direct Cartesian neuron table, however, grows rapidly with state and input precision, and a post-hoc table merely reproduces a previously trained recurrence. A useful lookup neuron must therefore preserve the temporal current contract that downstream LIF state updates expect, not merely return a table value.

Feasibility evidence and the failed direct route. Our starting point is a completed functional audit on QKFormer (Zhou et al. 2024). In this audit, every evaluated MultiStepLIF node was replaced by an 8-bit finite table over membrane state and input current. On CIFAR-100, native affine, normalization, and LIF replacement changed Acc@1 from 77.58% to 77.38% at $T = 1$ and from 81.02% to 80.80% at $T = 4$. A subsequent full-graph audit, which additionally table-addressed attention interactions, pooling, residual addition, and output reduction, produced drops of 0.02 and 0.12 percentage points. These observations show that finite transition tables can participate in an end-to-end spiking computation graph. They do *not* show that a learned table is better than an arithmetic neuron, that the table is compact, or that the mechanism transfers beyond the evaluated configurations. The LIF tables alone store approximately 16.98 MiB of FP32 values, making the current construction a fidelity baseline rather than the proposed method.

The first trainable-neuron proof of concept tested the most direct version of this idea: a 6-bit Dense LUT-IF transition table trained with the backbone frozen. That registered run is a clear negative result. On QKFormer CIFAR-100 at $T = 4$, the trained dense table reached 78.42% Acc@1 from an 81.02% teacher, worse than a post-hoc dense transition table at 78.96% and a quantized arithmetic LIF at 78.90%. Paired logit MSE also worsened substantially after training. Since all 35 requested LIF targets executed, this is not a coverage failure. It is evidence that directly learning the whole state-input transition couples soft training interpolation, hard inference addresses, spike thresholds, reset, and task loss too

tightly.

From transition learning to current-normalized lookup.

We therefore propose CNL-LUT-LIF, a current-normalized lookup LIF unit. The unit first obtains a replacement current from a support-aware lookup table, optionally indexed by semantic or state-derived addresses. It then applies a temporal-channel affine calibration that matches the lookup-current moments to the teacher-current moments observed during calibration. The normalized current finally enters an exportable fixed LIF transition table. In this design, lookup remains discrete at inference, but the learned or calibrated part is a low-dimensional current contract rather than the entire recurrent transition.

The completed main comparison matrix rejects the broad version of this thesis. Across retained QKFormer CIFAR-10/100 seed-42 checkpoints at $T = 1/4$, CNL-LUT-LIF passes the registered per-run gate only on CIFAR-100 $T = 4$. It improves the post-hoc all-LIF replacement from 79.55% to 79.77% Acc@1 in that setting, but ties near-lossless post-hoc lookup on CIFAR-10 and fails to repair a large 6-bit all-LIF drop on CIFAR-100 $T = 1$. The resulting thesis is therefore bounded: current normalization is a useful diagnostic and setting-specific repair, not a general lookup-neuron solution.

Contributions and current status. The manuscript is organized around four evidence-backed contributions:

- a negative boundary result showing that the initial Dense LUT-IF objective fails against matched post-hoc and quantized-arithmetic controls;
- a current-normalized lookup LIF formulation that decomposes lookup replacement into address, current-distribution, and LIF-transition errors;
- a fixed main comparison matrix with original LIF, post-hoc dense transition LUT, quantized arithmetic LIF, CNL-LUT-LIF without normalization, and CNL-LUT-LIF with normalization across CIFAR-10/100 and $T = 1/4$; and
- a claim boundary showing that CNL-LUT-LIF helps the CIFAR-100 $T = 4$ all-LIF replacement row, while CIFAR-10 is already near-lossless under post-hoc lookup and CIFAR-100 $T = 1$ exposes a precision or state-range failure.

Related Work

Spiking models and trainable neuron dynamics. Spiking vision models increasingly combine task-level learning with explicit temporal state. Spikformer established a Transformer-style SNN backbone, Spike-Driven Transformer removed many floating-point attention operations from the spiking path, SpikingResformer studied residual vision designs, and QKFormer introduced hierarchical Q-K attention for spiking visual recognition (Zhou et al. 2023; Yao et al. 2023; Shi, Hao, and Yu 2024; Zhou et al. 2024). Addition-only and saccadic spiking Transformers further show that the field is moving toward architectures whose internal operations are easier to map to discrete execution (Guo

et al. 2025; Wang et al. 2025). These works primarily innovate at the network, attention, or residual-block level. Parametric and temporal-learning approaches learn selected continuous or temporal neuron properties (Fang et al. 2020; Zhang and Li 2020). In contrast, CNL-LUT-LIF asks whether the discrete neuron-update interface itself can be replaced by calibrated lookup execution without changing the trained backbone.

Low-precision neuron updates. Low-precision SNNs may quantize membrane states, input currents, thresholds, spike timing, or attention arithmetic. Quantized Spike-Driven Transformer is a close example of preserving the spiking computation form while reducing numerical precision (Qiu et al. 2025), and conversion-based low-latency SNNs similarly show that quantized arithmetic is a serious alternative to lookup replacement (Hu, Tang, and Pan 2023). More broadly, binary and truth-table-oriented neural inference frameworks show why finite-precision signals are attractive for hardware mapping (Umuroglu et al. 2016, 2020). Such methods provide the closest arithmetic baseline for this paper because they retain an explicit recurrence and therefore separate “finite precision” from “lookup replacement.” We compare CNL-LUT-LIF with a quantized arithmetic LIF at the same state and input bit widths, rather than treating a full-precision LIF teacher as the only control.

Lookup-based neural computation. Lookup-oriented neural inference has been studied from several angles. LUT-Net maps neural computation onto FPGA soft logic (Wang et al. 2019); LUT-NN and LookupFFN replace expensive learned transformations with centroid or table lookup for efficient inference (Tang et al. 2023; Zeng et al. 2023); MADDNESS shows that table-based approximations can remove many multiplications from matrix products (Blalock and Gutttag 2021). Logic-gate networks and LUT-DNN systems further show that table and Boolean-function representations can be trained or synthesized as neural computation substrates (Petersen et al. 2022; Andronic and Constantinides 2024; Weng et al. 2025). Recent LUT-DNN work emphasizes that connectivity, fan-in, and function decomposition are central design variables rather than implementation details (Lou, Wu, and Leong 2025; Andronic and Constantinides 2023; Lou et al. 2024). Older SNN simulation work also used lookup tables to characterize neuronal dynamics (Ros et al. 2006). These lines make clear that lookup tables and lookup neurons are not new as a general device. The remaining question for this paper is narrower: whether a recurrent spiking LIF state interface can survive hard-address replacement under matched post-hoc and quantized-arithmetic controls. Look-up-table neurons for vision Transformers provide an especially relevant hardware-facing precedent (Nag et al. 2025); our paper differs by stress-testing a spiking LIF state interface with a negative dense-transition route and a mixed full-network gate.

What this paper does not claim. The paper does not claim that lookup tables are new as a general approximation device, nor that table execution alone implies compactness or hardware speed. The intended distinction is narrower: a task-

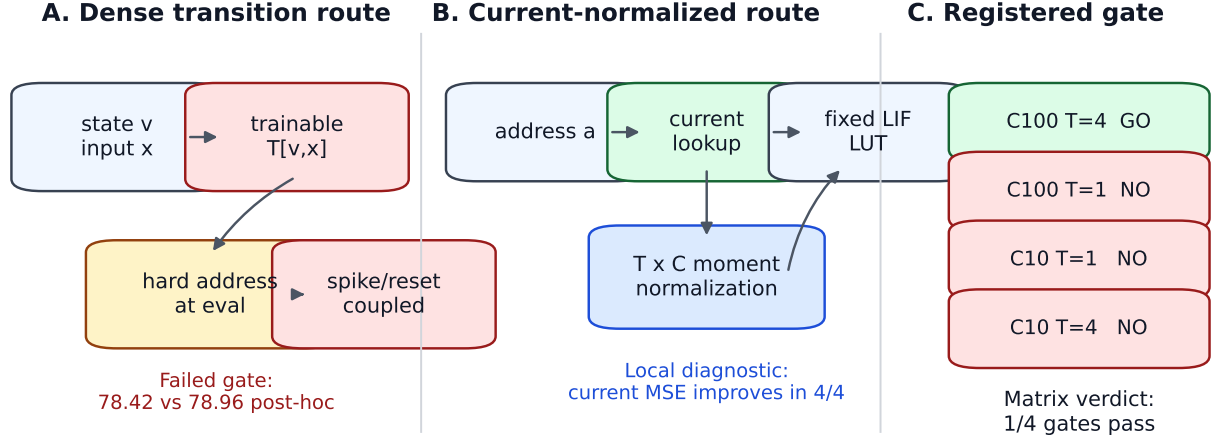


Figure 1: Evidence flow for the boundary study. Direct dense transition learning couples hard addresses, thresholding, and reset, and fails its matched-control gate. CNL-LUT-LIF instead calibrates lookup current before a fixed LIF transition table. The registered CIFAR-10/100, $T = 1/4$ matrix shows that current normalization improves the local current-MSE diagnostic in all rows but passes the full network gate only on CIFAR-100 $T = 4$.

trained spiking Transformer is evaluated under all-LIF neuron replacement, with a negative dense-transition training route, matched post-hoc transition tables, matched quantized arithmetic LIF, and a current-normalized lookup variant all measured under the same validation protocol. This framing makes negative and tied rows part of the evidence, rather than failures to be hidden.

Relation to QK-LUTFormer. QK-LUTFormer studies semantic Q/K response addresses, temporal-channel current calibration, and functional table execution. CNL-LUT-LIFuses these results as mechanism evidence for restoring lookup-current dynamics, while the finite-LIF audit remains feasibility evidence for exportable LIF state transitions. Q/K semantics are optional addresses for the current-lookup module, not an unrestricted claim that every spiking Transformer has the same lookup contract.

Method

Problem Formulation

Consider a time-major spiking layer with input current x_t , membrane state v_t , binary spike s_t , and next state v_{t+1} . A conventional discrete-time LIF neuron evaluates

$$\tilde{v}_t = v_t + \frac{x_t - v_t}{\tau}, \quad (1)$$

$$s_t = \mathbb{I}[\tilde{v}_t \geq \vartheta], \quad (2)$$

$$v_{t+1} = (1 - s_t)\tilde{v}_t + s_tv_{\text{reset}}, \quad (3)$$

where τ is the membrane time constant and ϑ is the threshold. Other common formulations change the placement of the input decay or reset but retain the same recurrent structure.

A post-hoc transition table discretizes v_t and x_t after the network has been trained. Let Q_v and Q_x denote state and

input quantizers. The post-hoc table stores

$$\mathcal{T}_{\text{post}}[Q_v(v), Q_x(x)] = f_{\text{LIF}}(v, x), \quad (4)$$

where f_{LIF} is the fixed recurrence. This representation can reproduce the neuron but does not correct a lookup current whose time-step or channel statistics no longer match the trained network.

Why the Dense Transition Route Is Not the Method

The first registered proof of concept replaced the fixed pre-threshold update with a trainable dense table,

$$a_t = (Q_v(v_t), Q_x(x_t)), \quad (5)$$

$$\tilde{v}_t = \mathcal{T}_\theta[a_t], \quad (6)$$

$$s_t = \mathbb{I}[\tilde{v}_t \geq \vartheta_\theta], \quad (7)$$

$$v_{t+1} = (1 - s_t)\tilde{v}_t + s_tv_{\text{reset}}. \quad (8)$$

For state grid $\{\bar{v}_i\}_{i=1}^{N_v}$ and input grid $\{\bar{x}_j\}_{j=1}^{N_x}$, the table was initialized by

$$\mathcal{T}_\theta^{(0)}[i, j] = \bar{v}_i + \frac{\bar{x}_j - \bar{v}_i}{\tau}. \quad (9)$$

Training used bilinear interpolation for gradients and hard nearest-neighbor addresses at evaluation. The backbone was frozen, and only transition values and per-module thresholds were optimized with

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\hat{g}, y) + \lambda_{\text{KD}}\mathcal{L}_{\text{KL}}(\hat{g}, g) + \lambda_\Delta \|\mathcal{T}_\theta - \mathcal{T}^{(0)}\|_2^2 + \lambda_{\text{nbr}}\mathcal{R}_{\text{nbr}}. \quad (10)$$

where $\mathcal{R}_{\text{nbr}}$ penalizes abrupt changes between adjacent entries. This route failed its preregistered gate and is therefore used as a boundary result, not as the proposed method. The failure motivates moving the adaptable component away from the whole recurrent transition.

The failure mode is structurally plausible. A transition table entry simultaneously determines the pre-threshold membrane value, the binary spike decision, and the reset path. During training, interpolation gives nearby entries shared gradients, but deployment uses hard integer addresses. A small movement of values around the threshold can therefore change spike timing discontinuously, while the frozen downstream network still expects the original current and spike-rate statistics. Direct dense transition learning optimizes a large table at the point where quantization, thresholding, reset, and task loss are most tightly coupled.

Current-Normalized Lookup LIF

CNL-LUT-LIF decomposes lookup replacement into three auditable modules:

$$\hat{x}_t = \mathcal{L}(a_t), \quad \tilde{x}_t = \mathcal{N}_{t,c}(\hat{x}_t), \quad (s_t, v_{t+1}) = \mathcal{T}_{\text{lif}}(v_t, \tilde{x}_t). \quad (11)$$

Here a_t is a discrete address, $\mathcal{L}$ is a support-aware current table, $\mathcal{N}_{t,c}$ is a temporal-channel current-normalization map, and $\mathcal{T}_{\text{lif}}$ is an exportable LIF transition table. The method hypothesis is that lookup errors are easier to control at the current level than inside the full spike/reset recurrence.

This decomposition changes the object of calibration. The current lookup table may be wrong because a discrete address is under-supported, because the address groups examples with different current magnitudes, or because the resulting current has a time/channel distribution that differs from the teacher. These errors can be measured before the spike threshold. The final LIF table then remains a fixed execution primitive with the same spike and reset semantics as the teacher recurrence, so any remaining failure can be attributed to current approximation, quantization range, or downstream sensitivity rather than to a moving recurrent rule.

Support-aware current lookup. Given address a , the table stores a calibration-set current prototype,

$$\mathcal{L}(a) = \frac{1}{|\mathcal{D}_a|} \sum_{(x,a') \in \mathcal{D}: a'=a} x. \quad (12)$$

If $n(a)$ is below a fixed support threshold, the module falls back to a coarser address or global current prototype. Every fallback is counted. The address may be a state/input code, a token/channel code, or a Q/K semantic code when the architecture exposes one; the paper does not assume that all backbones provide the same semantic address.

Temporal-channel current normalization. Lookup currents may have the correct address but the wrong scale or offset for the downstream spiking block. We estimate teacher-current and lookup-current moments on a calibration set:

$$\mu_{t,c}^{\text{ref}}, \sigma_{t,c}^{\text{ref}}, \mu_{t,c}^{\text{lut}}, \sigma_{t,c}^{\text{lut}}. \quad (13)$$

At inference, the lookup current is mapped back to the teacher-current coordinate system:

$$\tilde{x}_{t,c} = \frac{\hat{x}_{t,c} - \mu_{t,c}^{\text{lut}}}{\sigma_{t,c}^{\text{lut}} + \epsilon} \sigma_{t,c}^{\text{ref}} + \mu_{t,c}^{\text{ref}}. \quad (14)$$

An equivalent affine form $\tilde{x}_{t,c} = \gamma_{t,c}\hat{x}_{t,c} + \beta_{t,c}$ may be initialized from moments and lightly adapted under a fixed budget. Any adaptation is a separate ablation and may not be tuned after seeing validation performance.

Exportable LIF transition table. The normalized current enters a fixed LIF transition table,

$$\mathcal{T}_{\text{lif}}[Q_v(v_t), Q_x(\tilde{x}_t)] = f_{\text{LIF}}(v_t, \tilde{x}_t). \quad (15)$$

The spike and reset semantics remain those of the original LIF recurrence. This table is an execution primitive rather than the learned source of task gain. It is included because prior audits show that finite LIF transition tables can execute inside the evaluated QKFormer graph with small accuracy changes.

Diagnostics and Future Adaptation

The completed main comparison reports network accuracy, paired logit MSE, current MSE before and after normalization, spike-rate drift, fallback rate, and metadata-inclusive storage. No trainable adaptation is enabled in the reported matrix: the support-aware current table, temporal-channel moment normalization, and fixed LIF transition table are calibrated under the preregistered 128-batch budget and then evaluated on the complete validation split. A future adaptation ablation could restrict training to the current-normalization affine parameters and a small current residual correction:

$$\mathcal{L}_{\text{cnl}} = \mathcal{L}_{\text{CE}}(\hat{y}, y) + \lambda_{\text{KL}} \text{KL}(\hat{y}, y^{\text{teacher}}) + \lambda_x \|\tilde{x} - x^{\text{teacher}}\|_2^2 + \lambda_s \|r_s - r_s^{\text{teacher}}\|_2^2. \quad (16)$$

This objective is not used for the reported results and should not be cited as positive evidence.

Storage Accounting

Storage is treated as a claim-control variable rather than a success claim. When a storage number is reported, the accounting must include current prototypes, state/input grids or boundaries, LIF transition tables, thresholds, reset parameters, current-normalization moments or affine parameters, support counts, indices, and fallback state. Parameter count and storage are tracked separately from the frozen backbone because only the lookup neuron module is under study.

This accounting rule prevents a favorable accuracy row from being interpreted as a compact implementation when table values, ranges, support metadata, or normalization statistics dominate the footprint. The current paper therefore uses storage only as an audit and reproducibility quantity. Measured latency, energy, SRAM, and area require a separate hardware or simulator-backed study and are intentionally excluded from the success gate.

Experiments

The experiment section separates feasibility evidence, negative dense-LUT evidence, and the completed CNL-LUT-LIF comparison matrix.

Table 1: Registered dense-transition proof of concept. The trained dense table is a negative boundary result, not a CNL-LUT-LIF result.

Method	Acc@1	Drop	Logit MSE
Original LIF	81.02	0.00	—
Post-hoc dense LUT	78.96	2.06	24.5778
Quantized arithmetic LIF	78.90	2.12	24.5598
Dense LUT-IF init.	78.77	2.25	24.5471
Dense LUT-IF trained	78.42	2.60	56.6162

Completed Evidence

The completed QKFormer audits provide feasibility evidence only. Native affine, normalization, and LIF replacement produced paired accuracy drops of 0.20 and 0.22 points at $T = 1$ and $T = 4$. Complete functional lookup execution produced drops of 0.02 and 0.12 points. These frozen post-hoc rows are not attributed to CNL-LUT-LIF.

Prior current-replacement diagnostics also support the mechanism behind CNL-LUT-LIF. On QKFormer CIFAR-10, temporal-channel moment calibration changed static lookup-current replacement from 1.00/0.62-point drops at $T = 1/4$ to 0.07/−0.11 points. A two-stage current replacement at $T = 4$ recovered from an 88.99% static-hard row to 95.43% with moment calibration, a 0.27-point drop from the 95.70% teacher. These rows support current-contract restoration; they do not yet test all-LIF CNL-LUT-LIF replacement.

Negative Dense LUT-IF Proof of Concept

The first registered LUT-neuron proof of concept used QKFormer on CIFAR-100 at $T = 4$, seed 42. All 35 LIF nodes were replaced while the backbone was frozen. State and input used uniform 6-bit grids. Dense transition tables were initialized from the original LIF recurrence and trained for three epochs with a fixed batch budget. Evaluation used the complete 10,000-image validation split.

CNL-LUT-LIF Main Comparison Matrix

The main CNL-LUT-LIF matrix is fixed before execution as `cnl_lut_lif_main_matrix_seed42_20260628`. It asks whether current-normalized lookup LIF improves fixed 6-bit all-LIF replacement across retained QKFormer CIFAR-10/100 checkpoints at $T = 1/4$, all at seed 42.

The fixed controls are the original frozen LIF teacher, a 6-bit post-hoc dense transition LUT, a 6-bit quantized arithmetic LIF, CNL-LUT-LIF without current normalization, and CNL-LUT-LIF with temporal-channel current normalization. The CIFAR-100 $T = 4$ row also keeps the failed trained Dense LUT-IF proof of concept as a boundary reference. Every row uses 128 calibration batches and the complete validation split. No learning-rate, bit-width, checkpoint, stage, seed, or checkpoint-selection sweep is performed.

This design intentionally tests two different questions. The post-hoc dense transition LUT asks how much accuracy is lost by replacing the arithmetic recurrence with the same-precision hard table after training. The quantized

arithmetic LIF asks whether the same 6-bit state and input grids are already sufficient without lookup learning. The no-normalization CNL-LUT-LIF row isolates the effect of adding the support-aware current lookup path without restoring current statistics, while the normalized row tests the proposed temporal-channel moment correction. A CNL-LUT-LIF run is therefore counted as a success only if it executes all requested LIF targets, improves over post-hoc lookup on Acc@1 drop or paired logit MSE, has a smaller paired accuracy drop than the same-run post-hoc LUT, reduces current MSE relative to the no-normalization row, and stays below the dataset-specific drop threshold.

Reproducibility. The main matrix is tied to code snapshot `959782f`, the registered main-matrix run specification, and retained seed-42 QKFormer checkpoints. Each full row uses the complete 10,000-image validation split, batch size 32, 128 calibration batches, 6-bit state and 6-bit input grids, and all discovered MultiStepLIF targets. The anonymous artifact package should include the run specification, per-row metrics JSON, aggregate CSV, and the script that launches the fixed matrix; raw private paths and machine-specific usernames are excluded from the manuscript.

The matrix gate passes in one of four settings, matching the evidence flow in Fig. 1. CIFAR-100 $T = 4$ shows the intended effect: temporal-channel normalization improves the post-hoc all-LIF replacement from 79.55% to 79.77% and reduces the paired drop from 1.47 to 1.25 points. CIFAR-10 $T = 1/4$ is already near-lossless under post-hoc lookup, so CNL-LUT-LIF ties rather than improves the matched control. CIFAR-100 $T = 1$ is the most important failure in the fixed 6-bit matrix: all fixed all-LIF variants drop by more than six points.

Fixed 8-bit boundary diagnostic. To test whether the CIFAR-100 $T = 1$ failure is an uninterpreted negative row or a precision-sensitive interface boundary, we preregister one additional diagnostic and change only the transition state/input precision from 6 to 8 bits. The checkpoint, dataset, seed, calibration budget, targets, and full-validation protocol remain fixed. Table 3 shows that the paired drop decreases from 6.79 to 0.46 points for the post-hoc and CNL-LUT-LIF rows, passing the diagnostic gate. However, CNL-LUT-LIF still ties the same-run post-hoc LUT, so this result supports a precision/state-range boundary interpretation for CIFAR-100 $T = 1$ but not a broader CNL-LUT-LIF superiority claim.

Table 4 separates local and task-level evidence. The current-MSE diagnostic improves in every setting, confirming that the moment calibration performs the operation it was designed to perform. However, CIFAR-10 $T = 1$ improves paired logit fidelity only marginally while matching the post-hoc accuracy drop, and CIFAR-10 $T = 4$ slightly worsens paired logit MSE relative to post-hoc lookup despite keeping the same top-1 accuracy. The diagnostic is thus useful for explaining the mechanism but cannot be substituted for the registered network-level gate.

Experiment-Design Implications

The matrix is deliberately small but decision-oriented. It covers two datasets and two temporal depths for the same

Table 2: CNL-LUT-LIF main comparison matrix. Values are Acc@1 with paired drop in parentheses. Only CIFAR-100 $T = 4$ passes the registered per-run gate; the matrix does not support a broad superiority claim.

Setting	Teacher	Post-hoc LUT	Quantized LIF	CNL no norm	CNL norm
CIFAR-100 $T = 1$	77.58	70.79 (6.79)	70.87 (6.71)	70.79 (6.79)	70.79 (6.79)
CIFAR-100 $T = 4$	81.02	79.55 (1.47)	79.61 (1.41)	79.55 (1.47)	79.77 (1.25)
CIFAR-10 $T = 1$	94.90	94.53 (0.37)	94.43 (0.47)	94.53 (0.37)	94.53 (0.37)
CIFAR-10 $T = 4$	95.73	95.64 (0.09)	95.54 (0.19)	95.64 (0.09)	95.64 (0.09)

Table 3: CIFAR-100 $T = 1$ fixed 8-bit diagnostic. Values are Acc@1 with paired drop in parentheses. The row explains the severe 6-bit failure as precision-sensitive in this evaluated setting, but CNL-LUT-LIF ties post-hoc lookup at 8 bits.

Protocol	Post	Q-LIF	CNL-0	CNL
6-bit	70.79 (6.79)	70.87 (6.71)	70.79 (6.79)	70.79 (6.79)
8-bit	77.12 (0.46)	77.07 (0.51)	77.12 (0.46)	77.12 (0.46)

Table 4: Gate diagnostics for the normalized CNL-LUT-LIF row. Current normalization reduces current MSE in all rows, but this local diagnostic is not sufficient for the network-level gate.

Setting	Gate	Tgt.	Beats	MSE no	MSE norm
C100 $T = 1$	NO	yes	no	0.01471	0.01442
C100 $T = 4$	GO	yes	yes	0.02546	0.02469
C10 $T = 1$	NO	yes	yes	0.01522	0.01490
C10 $T = 4$	NO	yes	no	0.02401	0.02303

retained architecture, keeps the state/input precision fixed inside the main comparison, and evaluates all rows on the full validation split. This is sufficient to reject the broad claim that temporal-channel current normalization generally improves fixed 6-bit all-LIF replacement, because three of four registered rows do not pass. The single 8-bit follow-up diagnoses the worst failed row as precision-sensitive, but it is not a bit-width robustness study because no sweep or independent seed is run. Cross-architecture generality, seed stability, and hardware efficiency likewise require new preregistered experiments with their own matched controls.

After the Gate

Because the broad matrix gate fails, the next manuscript version should not claim CNL-LUT-LIF as a generally stronger lookup neuron. The defensible result is a boundary study: finite LIF lookup is executable, direct dense transition learning fails, and temporal-channel current normalization helps one high-temporal-reuse regime while leaving near-lossless and precision-limited regimes unchanged. Hardware latency, energy, SRAM, and deployment efficiency remain out of scope unless directly measured.

Discussion

Learning versus preserving dynamics. The central scientific distinction is between reproducing a fixed recurrence,

directly learning an unstable transition table, and preserving the current contract needed by the recurrent state update. A post-hoc table can be highly faithful without constituting a learning contribution. Conversely, an apparent accuracy gain is insufficient when it comes from extra capacity, soft interpolation at evaluation, or a weaker arithmetic baseline. The final analysis must therefore report hard-address inference, matched precision and storage, and a no-normalization ablation.

Local fidelity and network performance. A lower transition error need not produce a better classifier. Small recurrent differences may be suppressed by later layers, while a locally less faithful transition may improve the task objective. We therefore report transition diagnostics and task performance separately and analyze their relationship rather than assuming monotonic transfer.

Capacity and metadata. Dense state-input tables expose the main scaling risk. Value count alone understates practical storage when boundaries, indices, support counts, and fallback state are excluded. All compactness claims use metadata-inclusive accounting. A smaller table that loses the main empirical advantage is treated as a trade-off, not an unqualified improvement.

Architecture dependence. The pilot evidence originates from QKFormer, and the strongest current-normalization evidence comes from QK-style current replacement. Architecture-specific semantic context is optional, not a universal assumption. A favorable QKFormer result without a non-QK transfer setting supports an evaluated-configuration claim only.

Setting dependence. The main matrix shows that the normalized current path is not uniformly better than post-hoc lookup. It helps CIFAR-100 $T = 4$, ties already near-lossless CIFAR-10 rows, and does not repair CIFAR-100 $T = 1$ under the fixed 6-bit interface. The preregistered 8-bit diagnostic reduces the CIFAR-100 $T = 1$ drop from 6.79 to 0.46 points, so the severe 6-bit failure is precision/state-range sensitive in this evaluated setting. This does not make CNL-LUT-LIF broadly superior: at 8 bits, CNL-LUT-LIF still ties same-run post-hoc lookup. The most plausible interpretation is that temporal-channel normalization helps when lookup quantization perturbs a reusable dynamic current contract, while the worst $T = 1$ row is dominated by transition-interface precision.

Why a negative matrix can still be useful. For a lookup-neuron paper, a broad positive claim would be attractive but

unsupported. The main result instead identifies which part of the design is reliable. Finite LIF tables execute, current normalization reduces measured current mismatch, and one high-temporal-reuse row gains network accuracy over post-hoc lookup. At the same time, near-lossless rows reveal a ceiling where there is little room to improve, and CIFAR-100 $T = 1$ reveals a precision boundary where the fixed interface is too lossy. This is actionable information for the next design: improve the quantized state/input interface or add a separately registered adaptive current model before claiming general replacement.

Limitations. The completed dense-transition proof of concept is negative, and the CNL-LUT-LIF matrix is mixed. The current evidence does not establish a generally superior all-LIF replacement, adaptive coding, factorized storage, broad transfer, or end-to-end co-training. Hardware latency, energy, SRAM, and deployment efficiency are outside the scope of this paper.

Conclusion

We evaluated CNL-LUT-LIF, a current-normalized lookup LIF unit that separates lookup-current quantization, temporal-channel current calibration, and exportable LIF transition lookup. The evidence supports three bounded conclusions. First, finite LIF tables can execute inside the evaluated QKFormer graph. Second, the first direct Dense LUT-LIF training route fails against matched controls. Third, current normalization improves the CIFAR-100 $T = 4$ all-LIF replacement row but does not yield a broad win across the CIFAR-10/100 $T = 1/4$ matrix. The resulting paper should be framed as a boundary and diagnostic study rather than a general solved lookup-neuron method. Hardware and deployment claims are outside the scope of the paper.

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